

APPLIED PHYSICS

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5BS08	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	30	70	100
Contact Classes:44	Tutorial Classes:8	Practical Classes: 0				Total Classes: 52		
COURSE OBJECTIVES The course should enable the students to: 1. Learn the behavior of matter waves and applications of Schrödinger wave equations 2. Understand the formation of energy bands in solids 3. Acquire the fundamental knowledge of semiconductor devices and their applications 4. Learn the basic principles of laser and optical fiber 5. Describe the fundamentals in quantum computations and explain how it can be used in cryptography								
COURSE OUTCOMES At the end of the course students will be able to: 1. Apply principles of quantum mechanics to calculate observables on known wave functions. 2. Apply the concept of band theory to explain the behavior of different electronic materials. 3. Analyze the inner working of semiconductor p-n diodes, LED and new semiconductor devices. 4. Explain the basic principles of laser physics, working of lasers, principle of propagation of light in Optical fibers and Solve the numerical problems associated 5. Apply the basic tools and techniques of quantum computation and quantum information in a problem solving scenario								
UNIT - I	INTRODUCTION TO QUANTUM PHYSICS						CLASSES: 10	
Black body radiation, Planck's law, photoelectric effect, Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.								
UNIT - II	BAND THEORY OF SOLIDS						CLASSES: 10	
Band theory - Quantum theory of free electron, Bloch theorem, Kronig-Penny model, E-k diagram, effective mass of electron. Energy bands in Solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac Statistics (Qualitative treatment), classification of materials as conductors, insulators and semiconductors.								
UNIT - III	SEMI-CONDUCTORS						CLASSES:12	
Semiconductor Physics: Intrinsic and Extrinsic Semiconductors – formation of p-type and n-type, Fermi Level. Direct and Indirect Band gap semiconductors, Hall Effect and applications. Physics of Semiconductor Devices: PN Junction Diode, V-I characteristics and applications. LED - Construction, working and applications. Solar cells- working and its applications. Efficiency issues of Solar cell.								
UNIT - IV	LASER & FIBER OPTICS						CLASSES: 10	
Lasers: Characteristics of Laser, Basic processes between two energy levels. Pumping mechanism, Meta stable state and Population inversion. Working of Nd-YAG laser, applications of lasers in different fields.								

Fiber Optics: Structure of fibers, Principle of fiber (TIR), Acceptance angle and NA. Types of fibers- SI and GI fibers- R.I profiles. Single and Multimode fibers-SMSI, MMSI, MMGI. OFC System with block diagram. Fiber optic sensors – Basic principle, working of Pressure and Temperature Sensors. Applications of fibers in different fields.

UNIT - V**QUANTUM COMPUTATION AND CRYPTOGRAPHY****CLASSES: 10**

Quantum computation: Idea of classical bits and qubits, Bloch vector representation of state of qubit. Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form. Two level gates: CNOT and SWAP gates. Comments on No cloning theorem. Quantum Teleportation – Basic Idea.

Cryptography Methods: Introduction to cryptography, Classical and Public key cryptosystems, Verman cipher, The RSA protocol; Quantum Key distribution protocol - BB84 protocol.

TEXT BOOKS

1. Engineering Physics, B.K. Pandey, S. Chaturvedi – Cengage Learning
2. R. Robinett, "Quantum Mechanics", OUP Oxford, 2006.11ndEdn.
3. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVthEd
4. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press

REFERENCE BOOKS

1. J. Singh, Semiconductor Optoelectronics: Physics and Technology, Mc Graw-Hill inc.(1995)
2. O. Svelto, "Principles of Lasers", Springer Science & Business Media, 2010
3. D. A. Neamen, "Semiconductor Physics and Devices", Times Mirror High Education Group, Chicago, 1997
4. Principles of Quantum computation and Information – By G. Benenti, G. Casati, G. Strini, World Scientific

WEB REFERENCES

1. https://www.edx.org/course?search_query=semiconductor+physics
2. <https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x>

E-TEXT BOOKS

1. http://www.phys.sinica.edu.tw/TIGP-ANO/Course/2010_Fall/classnotes/NanoB_week14.pdf
2. <https://www.scribd.com/document/70908178/Semiconductor-Devices-Basic-Principles-Jasprit-Singh>
3. <https://www.scribd.com/doc/105174065/Fundamentals-of-Photonics>
4. [ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20\(2ed.,%20GTP,%20Springer,%202010\)\(ISBN%20144196441X\)\(O\)\(674s\)_PEo_.pdf](ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20(2ed.,%20GTP,%20Springer,%202010)(ISBN%20144196441X)(O)(674s)_PEo_.pdf)
5. https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf

MOOCS COURSE

1. <http://nptel.ac.in/courses/115102025/> (Fundamental concepts of semiconductors)
2. <http://nptel.ac.in/courses/104104085/2> (Lasers and its applications)
3. <https://nptel.ac.in/courses/115/101/115101092/> (Quantum computing)